

Tutorial - 2

$$1(a) \quad -\frac{6}{\sqrt{40}} \hat{r} - \frac{38}{\sqrt{40}} \hat{\phi}$$

$$(b) \quad \vec{v}_t = 0.043 \hat{\theta} + 100 \hat{\phi}$$

$$\hat{n} = -0.9988 \hat{r} - 0.0499 \hat{\phi}$$

Griffiths:

$$1.42 (a) \quad \vec{\nabla} \cdot \vec{v} = \frac{1}{r} \frac{\partial}{\partial r} [r r (2 + \sin^2 \phi)] + \frac{1}{r} \frac{\partial}{\partial \phi} (r \sin \phi \cos \phi) + \frac{\partial}{\partial z} (3z) = 8$$

$$(b) \quad \int (\vec{\nabla} \cdot \vec{v}) d\tau = \int 8 d\tau = 8 \int_{r=0}^2 \int_{\phi=0}^{\pi/2} \int_0^5 r dr d\phi dz$$

$$= 40\pi$$

$$\oint \vec{v} \cdot d\vec{a} = \left[\int_1 + \int_2 + \int_3 + \int_4 + \int_5 \right] \vec{v} \cdot d\vec{a}$$

$$\text{Surface (1): } z=0, r: 0-2, \phi: 0-\pi/2$$

$$d\vec{a} = -r dr d\phi \hat{z}$$

$$\vec{v} \cdot d\vec{a} = -3z r dr d\phi = 0$$

$$\text{Surface (4): } z=5, r: 0-2, \phi: 0-\pi/2$$

$$d\vec{a} = r dr d\phi \hat{z}$$

$$\vec{v} \cdot d\vec{a} = 3z r dr d\phi$$

$$\therefore \int_4 \vec{v} \cdot d\vec{a} = 15 \int_0^2 r dr \int_0^{\pi/2} d\phi = 15\pi$$

$$\text{Surface (3): } r: 0-2, \phi=0, z: 0-5; d\vec{a} = -dr dz \hat{\phi}$$

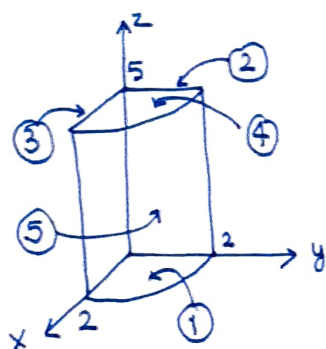
$$\vec{v} \cdot d\vec{a} = -r \sin \phi \cos \phi dr dz = 0$$

$$\text{Surface (2): } r: 0-2; \phi=\pi/2, z: 0-5; d\vec{a} = dr dz \hat{\phi}$$

$$\vec{v} \cdot d\vec{a} = r \sin \phi \cos \phi dr dz = 0$$

$$\text{Surface (5): } r=2, \phi: 0-\pi/2; z=0-5 d\vec{a} = r d\phi dz \hat{r}$$

$$\int_5 \vec{v} \cdot d\vec{a} = \int_0^{\pi/2} \int_0^5 r^2 (2 + \sin^2 \phi) d\phi dz = 25\pi$$



$$\Rightarrow \oint \vec{V} \cdot d\vec{a} = 40\pi$$

$$(c) \quad \text{curl}(\vec{V}) = 0$$

$$1.53 \quad \int_V (\nabla \cdot \vec{A}) d\tau = \oint_S \vec{A} \cdot d\vec{a}$$

$$\nabla \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \xi_1} (V_1 h_2 h_3) + \frac{\partial}{\partial \xi_2} (V_2 h_3 h_1) + \frac{\partial}{\partial \xi_3} (V_3 h_1 h_2) \right]$$

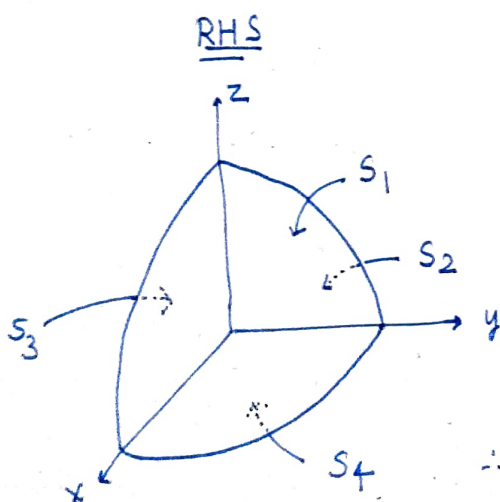
$$= 4r \sin \theta$$

$$\underline{\text{LHS}}: \int_V 4r \sin \theta d\tau = \iiint 4r \sin \theta r^2 \sin \theta dr d\theta d\phi$$

For 1 octant of the sphere of radius R
 $r = 0 - R$; $\theta = 0 - \pi/2$; $\phi = 0 - \pi/2$

$$\therefore \text{LHS} = 4 \int_0^R \int_0^{\pi/2} \int_0^{\pi/2} r^2 \sin \theta \cos \theta dr d\theta d\phi$$

$$= \frac{\pi R^4}{4}$$



$$\oint_S \vec{V} \cdot d\vec{a} = \left(\int_{S_1} + \int_{S_2} + \int_{S_3} + \int_{S_4} \right) \vec{V} \cdot d\vec{a}$$

Surface(1): $r=R$; $\theta: 0 - \pi/2$; $\phi: 0 - \pi/2$

$$d\vec{a} = R^2 \sin \theta d\theta d\phi \hat{r}$$

$$\vec{V} \cdot d\vec{a} = R^2 \cos \theta R^2 \sin \theta d\theta d\phi$$

$$\therefore \int_{S_1} \vec{V} \cdot d\vec{a} = R^4 \int_0^{\pi/2} \int_0^{\pi/2} \sin \theta \cos \theta d\theta d\phi$$

$$= \pi R^4 / 4$$

Surface(2): $r: 0 - R$; $\theta: 0 - \pi/2$; $\phi = \pi/2$

$$d\vec{a} = r dr d\theta \hat{\phi} \quad \vec{V} \cdot d\vec{a} = (-r^2 \cos \theta \sin \phi) r dr d\theta$$

$$= -r^3 \cos \theta d\theta dr$$

$$\therefore \int_{S_2} \vec{V} \cdot d\vec{a} = - \int_0^{\pi/2} \int_0^R r^3 \cos \theta dr d\theta = -R^4 / 4$$

Surface (3): $r: 0-R; \theta: 0-\pi/2; \phi=0$

$$d\vec{a} = -r dr d\theta \hat{\phi} \quad \vec{v} \cdot d\vec{a} = (r^2 \cos \theta \sin \phi) r dr d\theta = 0$$

Surface (4): $r: 0-R; \theta = \pi/2, \phi: 0-\pi/2$

$$d\vec{a} = r \sin \theta dr d\phi \hat{\theta}$$

$$\vec{v} \cdot d\vec{a} = r^2 \cos \phi r \sin \theta dr d\phi = r^3 \cos \phi dr d\phi$$

$$\int_{S_4} \vec{v} \cdot d\vec{a} = \int_0^R \int_0^{\pi/2} r^3 \cos \phi dr d\phi = R^4/4$$

$$\therefore \oint_S \vec{v} \cdot d\vec{a} = \pi R^4/4 = \text{LHS.}$$

1.58

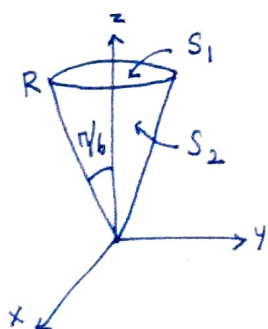
$$\vec{\nabla} \cdot \vec{v} = \frac{4r \cos^2 \theta}{\sin \theta}$$

For the volume of the 'ice-cream cone',

$$r \equiv 0-R, \quad \theta \equiv 0-\pi/6, \quad \phi \equiv 0-2\pi$$

$$\begin{aligned} \therefore \int (\vec{\nabla} \cdot \vec{v}) d\tau &= \int_0^R \int_0^{\pi/6} \int_0^{2\pi} \frac{4r \cos^2 \theta}{\sin \theta} r^2 \sin \theta dr d\theta d\phi \\ &= \int_0^R 4r^3 dr \int_0^{\pi/6} \cos^2 \theta d\theta \int_0^{2\pi} d\phi \\ &= \frac{\pi R^4}{12} (2\pi + 3\sqrt{3}) \end{aligned}$$

$$\oint \vec{v} \cdot d\vec{a} = \left[\int_{S_1} + \int_{S_2} \right] \vec{v} \cdot d\vec{a}$$



Surface 1: $r \equiv R, \theta: 0-\pi/6; \phi \equiv 0-2\pi$

$$d\vec{a} = R^2 \sin \theta d\theta d\phi \hat{r}$$

$$\vec{v} \cdot d\vec{a} = R^2 \sin \theta R^2 \sin \theta d\theta d\phi$$

$$\begin{aligned} \int_{S_1} \vec{v} \cdot d\vec{a} &= R^4 \int_0^{\pi/6} \sin^2 \theta d\theta \int_0^{2\pi} d\phi \\ &= \frac{\pi R^4}{12} (2\pi - 3\sqrt{3}) \end{aligned}$$

surface (2): $r: 0 \rightarrow R, \theta = \pi/6; \phi: 0 \rightarrow 2\pi$

$$d\vec{a} = r \sin \theta d\phi dr \hat{\theta} \\ = \frac{\sqrt{3}}{2} r d\phi dr \hat{\theta}$$

$$\vec{V} \cdot d\vec{a} = \sqrt{3} r^3 d\phi dr$$

$$\therefore \int_{S_2} \vec{V} \cdot d\vec{a} = \sqrt{3} \int_0^R r^3 dr \int_0^{2\pi} d\phi = \frac{6\sqrt{3}}{12} \pi R^4$$

1.33 Do as discussed in class-room.

- 1.43 (a) 20
(b) -1
(c) 0
(d) 0

1.55 Dropped